

## ANIMAL HUSBANDRY

### Last five-year previous year question 2019-2023

**1. Which one of following is best milk yield cow breed?**

- a. Holstein Friesian      b. Jersey      c. Sindhi      d. All of the above

**2. Finest carpet wool is obtained from which sheep breed?**

- a. Chokla sheep      b. Nellore      c. Pugal      d. Marwari

**3. Haryana is a famous breed of which domesticated animal**

- a. Sheep      b. Buffalo      c. Goat      d. Cow

**4. Which of the following is legume fodder**

- a. Sorghum      b. Pearl millet      c. Alfa lafa      d. All of the above

**5. Male sheep used for breeding are called**

- a. Spayed  
b. Wedder  
c. Ram  
d. Buck

**6. What is gestation period of sheep?**

- a. 170 days  
b. 197 days  
c. 157 days  
d. 100 days

**7. If a Ram is to be sold, its castration should be done when lambs are between-----  
of age**

- a. 5 weeks  
b. 10 weeks  
c. 2 weeks  
d. 15 weeks

- 8. The first yellow coloured milk produced by the cow after the child birth is to be fed to the baby for enhancing the immunity of the baby is called as**
- a. Colostrum
  - b. Weaning
  - c. Oxytocin
  - d. None of the above
- 9. The method where two breeds are crossed alternatively and is proposed for utilizing heterosis in both dams and progeny is \_\_\_\_\_**
- a. Mating
  - b. Inbreeding
  - c. **Crossbreeding**
  - d. Insemination
- 10. \_\_\_\_\_ is the cyclic period when the female cow is active for the mating with bull is called as a**
- a. Heat period
  - b. Estrous cycle
  - c. Mensuration
  - d. None of the above
- 11. Young female in the sheep is called \_\_\_\_\_.**
- a. Doe
  - b. Spayed
  - c. Ewe
  - d. Boar
- 12. \_\_\_\_\_ in animals is a series of recurring physiological changes in the female reproductive system that prepares them for mating and pregnancy.**
- a. Estrous cycle
  - b. Estrogen cycle
  - c. CAC cycle
  - d. None of the above
- 13. Castrated female of pig is called as?**
- A. Suckling

- B. Grunting
- C. Tordi
- D. Barrow

**14. The cow breed of Maharashtra has milch purpose or sometime used for draught also are known as?**

- A. Surti
- B. Kankrej
- C. Dangi
- D. Jafrabadi

**15. The process in which the calf is separated from mother called as\_\_\_\_\_**

- a. Castration
- b. Weaning
- c. Dehorning
- d. All of the above

**16. What is the gestation period of swine?**

- a. 100 days
- b. 150 days
- c. 114 days
- d. 137 days

**17. Which famous breed of buffalo has an special feature of wall eyes.**

- a. Murrah
- b. Nalli ravi
- c. Changtani
- d. Bhadawari

**18. What is the gestation period of indigenous cow breed?**

- a. 350 to 400 days
- b. 500 days
- c. 280- 290 days
- d. 200 – 250 days

**19. Which Indian sheep breed is also called as Indian merino ?**

- a. Shekhawati

- b. Bikaneri
- c. Marwari
- d. Jaffrabadi

**20. New born baby of the sheep is called as \_\_\_\_\_**

- A. Doe
- B. Ewe
- C. Lamb
- D. Spayed

**21. Jamnapuri is an important breed of \_\_\_\_\_ which is generally domesticated in UP.**

- A. Goat
- b. Cow
- c. Pig
- d. Sheep

**22. Adult male of goat is called as \_\_\_\_\_**

- A. Bull
- b. Boar
- c. Barrow
- d. Buck

**23. What is the minimum amount of floor space required for the young cows**

- A.  $2\text{m}^2$
- b.  $5\text{m}^2$
- c.  $10\text{m}^2$
- d.  $1\text{m}^2$

**24. What is the minimum space required for the domestication of Ram is in  $\text{m}^2$ ?**

- A. 3.2
- b. 5.2
- c. 7
- d. 10

**25. Goat meat is called as \_\_\_\_\_**

- A. Beef
- b. Chevon
- c. Pork
- d. Mutton

**26. Cow milk contains \_\_\_\_\_ percentage of SNF solid not fat**

- A. 10%
- b. 5%
- c. 8.5%
- d. 15%

**27. What should be the age of weaning of a calf?**

- A. 100 days
- b. 90 days
- c. 115 days
- d. 200 days

**Passage based question were asked from Swine breeding of 8 marks which contains the following question (MAINS 2022)**

**28. The act of mating of pig is called as \_\_\_\_\_**

- A. Coupling      b. Farrowing      c. Mating      d. Breeding

**29. The act of giving birth by the sow to the piglets is called as a \_\_\_\_\_**

- A. Farrowing      b. Coupling      c. Weaning      d. None of the above

**30. What is the gestation period of pig ?**

- A. 134 days      b 150 days      c. 115 days      d.170 days

**31 . How many times a pig farrowed in a year**

- a. 2 times      b. Only one time      c. 5 times      d. 3 times

## Answers

|       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A  | 2. A  | 3. D  | 4. C  | 5. C  | 6. C  | 7. C  | 8. A  | 9. C  | 10. A |
| 11. C | 12. A | 13. D | 14. C | 15. B | 16. C | 17. B | 18. C | 19. A | 20. C |
| 21. D | 22. D | 23. D | 24. A | 25. B | 26. C | 27. B | 28. A | 29. A | 30. C |
| 31. A |       |       |       |       |       |       |       |       |       |

## ANIMAL HUSBANDRY

Animal husbandry, agriculture and allied activities have been the **core livelihood for majority of the rural people** since time immemorial.

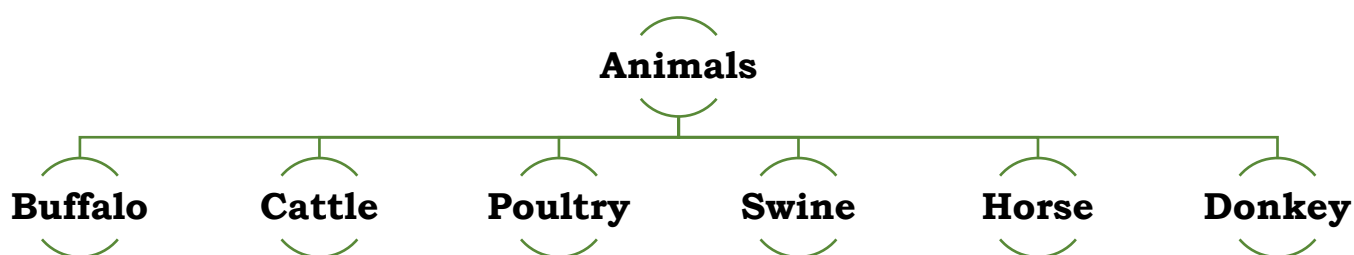
- It provides **productive employment, especially self-employment** and the most valuable supplementary income to a vast majority of rural households, majority of who are small and marginal farmers and landless laborers.
- Livestock provide increased economic stability to the poor masses. They **act as cash buffer in case of small stock and as captive reserve** in case of larger stock.
- Livestock **provide quality animal protein to human population in the form of milk, eggs, meat and value-added products.**
- **They provide draught power for agricultural operations**, organic manure for agriculture and raw materials like skin, hides, blood, bone, hoof, horn, etc., for various industries.

### Role Of Livestock in Human Life.

- **Provides food like milk, meat and other value-added products**, which helps in eradicating poverty and to have self-sufficiency in food security.
- **Provides valuable organic manure for agriculture**, thus improving soil fertility and productivity.
- **Provides draught animal power for different agricultural operations** and for transportation, and also to conserve fossil fuel.
- **Provides a platform for ensuring sustainability (biological and economic)**, which bringing about a change in the social status (society).
- **Provides skin and hides for leather products**, thus helping to make additional income and earn foreign exchange.
- **Provide other byproducts like hoof and horns**, there by promoting rural handicraft industries.
- **Provides wool and other natural fibers of importance**, thus helping to improve income and promote industrial growth.
- **Provides blood as a byproduct**, may help to produce hematinic and other pharmacological products.

- Provides **gainful rural employment for many people**, and to have full-time self-employment.
- **Provides a means of storing wealthy and** also a way to meet social obligations, thus helping to have security and overcome difficult times.
- **Provides a means of controlling weeds by grazing**, which helps to keep the premises clean.

## Animals domesticated under animal husbandry in India



## LIVESTOCK PRODUCTION AND MANAGEMENT

### Scientific Name Of Various Animals

| <u>Animal</u>                      | <u>Scientific Name</u>     |
|------------------------------------|----------------------------|
| <b>Zebu cattle</b>                 | <i>Bos indicus</i>         |
| <b>Exotic cattle</b>               | <i>Bos Taurus</i>          |
| <b>Goat</b>                        | <i>Capra hircus</i>        |
| <b>Sheep</b>                       | <i>Ovis aries</i>          |
| <b>River buffalo/Water buffalo</b> | <i>Bubalus bubalis</i>     |
| <b>Swamp buffalo</b>               | <i>Bubalus carabanesis</i> |
| <b>Pig</b>                         | <i>Sus domesticus</i>      |
| <b>Horse</b>                       | <i>Equus caballus</i>      |
| <b>Donkey</b>                      | <i>Equus asinus</i>        |
| <b>Chicken</b>                     | <i>Gallus domesticus</i>   |
| <b>Camel (single humped)</b>       | <i>Camelus dromedaries</i> |

**Livestock** comprises of all farm animals having economic importance (milk/meat/wool/draught purpose) Ex. Cattle, Buffalo, Sheep, Goat, Camel, Pig, Horse, Mule, Donkey, Pony etc.

## Terminologies Related To Various Domestic Animals

| <u>Animal</u> | <u>Adult male</u> | <u>Adult female</u> | <u>Group of animals</u> | <u>Act of mating</u> | <u>Act of parturition</u> | <u>New born</u> | <u>Young male</u> | <u>Young female</u> |
|---------------|-------------------|---------------------|-------------------------|----------------------|---------------------------|-----------------|-------------------|---------------------|
| <b>Cattle</b> | Bull              | Cow                 | Herd                    | <b>Serving</b>       | Calving                   | Calf            | Bull calf         | Heifer calf         |
| <b>Sheep</b>  | Ram               | <b>Ewe</b>          | Flock                   | Tupping              | Lambing                   | Lamb            | <b>Ram lamb</b>   | Gimmer lamb         |
| <b>Goat</b>   | Buck              | Doe/nanny           | Trip or band            | Serving              | Kidding                   | Kid             | Buckling          | Gloating            |
| <b>Pig</b>    | Boar              | Sow                 | Drove                   | <b>Coupling</b>      | <b>Farrowing</b>          | Piglet          | Boarling          | Gilt                |
| <b>Horse</b>  | Stallion          | Mare                |                         | Covering             | Foaling                   | Foal            | Colt              | Filly               |
| <b>Camel</b>  | Maiya/oont        | Sand                | Tola                    | Lakhana              | Calving                   |                 | tordia            | Tordi               |

**\*\*Most important table for NABARD**

## Other Important Terminologies

| <b>Animal</b>          | <b>House</b>      | <b>Castrated male</b>   | <b>Sound</b> | <b>Female with offspring</b> | <b>Castrated female</b> |
|------------------------|-------------------|-------------------------|--------------|------------------------------|-------------------------|
| <b>Cattle, Buffalo</b> | <b>Barn/ Byre</b> | Bullock / Steer         | Bellowing    | Calf-at-foot                 | <b>Spayed</b>           |
| <b>Sheep, goat</b>     | Pen               | <b>Wedder / Wether</b>  | Bleating     | Suckling                     | Spayed                  |
| <b>Pig</b>             | Sty               | <b>Hog/Stag/ Barrow</b> | Grunting     | Suckling                     | <b>Spayed</b>           |
| <b>Horse</b>           | Stable            | Geld / Gelding          | Neighing     |                              | Spayed                  |



- **Bullock**: Castrated bull for **draught purpose**
- **Steer**: Castrated bull for **meat purpose**
- **Teaser bull: Vasectomised bull** (by cutting& ligating the vas deferens) used for heat detection& teaser bull is not castrated.
- **Heifer**: A **female cow above 1 year**, which has not yet calved
- **Buller cow/Nymphomaniac cow**: A cow remains always in heat.
- **Crit /Card/Runt**: **Smallest piglet** in a litter usually last to be farrowed.
- **Castrated male** – Male animal made **infertile by removing its testicles** using surgical means.
- **Castrated female** – castrated females are **those whose reproductive organs are removed to make infertile**

## BUFFALO DOMESTICATION

The buffalo species originated in India. The buffaloes are normally **classified into river ad swamp types though both are called *Bubalus bublis***.

Indian buffaloes are in **important source of milk supply today** and yield nearly three times as much milk as cows. **More than half of the total milk produced (55%) in the country was contributed by the 47.22 million milch buffaloes**, whereas the 57.0 million cows contribute only 45% of the total milk yield. Indian Buffaloes are water buffaloes.

### Buffalo Breeds

#### Classification Of Buffalo Breed Based On Region

| <u>Murrah group</u> | <u>Gujarat group</u> | <u>UP group</u>  | <u>South India group</u> | <u>Central India group</u> |
|---------------------|----------------------|------------------|--------------------------|----------------------------|
| Murrah              | Surti                | <b>Bhadawari</b> | Toda                     | Nagpuri                    |
| <b>Nili Ravi</b>    | <b>Jaffarabadi</b>   | Tarai            | South Kanara             | Jerangi                    |
| Kundi               | Mehsana              |                  |                          | Kalahandi                  |
| Godavari            |                      |                  |                          | Sambalpur                  |
|                     |                      |                  |                          | <b>Pandharpuri</b>         |
|                     |                      |                  |                          | Manda                      |

## Description Of Buffalo Breeds

### Murrah

- It is the most important breed of buffaloes **whose home is Rohtak, Hisar and Jind** of Haryana and Nabha and Patiala districts of Punjab.
- **Synonyms:** Delhi, Kundi, Kali
- The colour is **usually jet black** with **white markings** on tail and face and extremities sometimes found.
- The **tightly curved horn** is an important character of this breed.
- The body **size is massive, neck and head are comparatively** long.
- Head of **females is short, fine** and clear cut.
- Hips are broad and fore and hind quarters are drooping.
- The buffalo of this breed are one of the **most efficient milk and butter fat producers** in India.
- Butter fat **content is 7% Average lactation** yield is varied **from 1500-2500 kg** the average **milk yield is 6.8 kg/day**.
- While a few individual animals yield much as 19.1 kg/day.
- Age at **first calving is 45-50 months** and inter **calving period is 450-500 days**.



**Figure 1 - Murrah breed**

### Nili Ravi

- This breed is found in Sutlej valley in Ferozpur district of Punjab **and in the Sahiwal district of Pakistan**.
- Usually, **the colour is black with white marking on forehead, face, muzzle, legs and tail**.
- The most desired character of the **female is the possession of white markings**.
- The **head is elongate**, bulging at top and depressed between eyes. The muzzle is fine.
- The **frame is medium sized**.
- **The peculiarity of the breed is the wall eyes**.

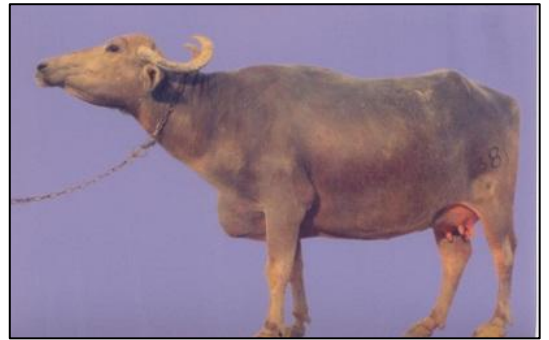


**Figure 2- Nilli Ravi breed**

- The **horns are small and coiled tightly**. The neck is long, thin and fine.
- The **milk yield is 1500-1850 kg per lactation** and the inter calving period is 500-550 days.
- Age at first **calving is 45-50 months**.

## Bhadawari

- **Agra and Etawah district of Uttar Pradesh and Gwalior district of Madhya Pradesh.**
- The body is **medium size and wedge-shaped horn**. The head is **comparatively small**, the legs are **short and stout**, and the hooves are black. The hind quarters are uniform and higher than the forequarter.
- The **body is usually light or copper coloured** is a peculiarity of this breed. Eye lids are generally copper or light brown colour.
- Two white lines 'Chevron' are present at the lower side of the neck similar to that of surti buffaloes.
- **Horns are black, curling slightly outward, downward before running backward parallel and close to neck, and finally turning upward.**
- The average **milk yield is 800 to 1000 kg**.
- The bullocks are good draught animal with high heat tolerance.
- **The fat content varies from 6 to 12.5 per cent**. This breed is an efficient converter of coarse feed into butterfat and is known for its high butter fat content.



**Figure 3 – Bhadawari**

## Jaffarabadi

- These are **massive animals found in their pure form in Gir forests**. The breeding tract of this breed is Kutch, and Jamnagar districts of Gujarat.
- The **head and neck are massive**. The forehead is very prominent, wide with a slight depression in the middle.



**Figure 4 – Jaffarabadi Breed**

- The **horns are heavy, inclined to droop at each side of the neck** and then turning up at point, but less tightly curved **than in Murrah (drooping horns)**.
- The colour is usually black.
- The **average milk yield is 1000 to 1200 kg**. These animals are mostly maintained by **traditional breeders called Maldharis, who are nomads**.
- The bullocks are heavy and **used for ploughing and carting**.

## Surati

- The breeding tract of this breed is **Kaira and Baroda district of Gujarat**.
- Coat colour varies from **rusty brown to silver-grey**. Skin is black or brown.
- The body is well shaped and medium sized; the barrel is wedge shaped.
- The head is long with prominent eyes.
- The **horns are sickle shaped**, moderately long and flat.
- The **milk yield ranges from 900 to 1300 kg**.
- The age at first calving is **40-50 months** with an inter-calving **period of 400-500 days**.
- The peculiarity of this **breed is very high fat percentage in milk (8-12 per cent)**.



**Figure 5 – Surati breed**

## Mehsana

- Mehsana is a **dairy breed of buffalo found in Mehsana town in Gujarat** and adjoining Maharashtra state.
- **Body is mostly black**; a few animals are black-brown in colour.
- The breed is supposed to have been **evolved out of crossbreeding between the Surati and the Murrah**.
- The **body is longer than in Murrah** and the limbs lighter.
- The head is **longer and heavier**.



**Figure 6 – Mehsana Breed**

- The horns **usually are less curved** at the end compared to Murrah breed but are longer and could be of irregular shape.
- The **milk yield is 1200-1500 kg**.
- The breed is **supposed to have good persistency**.
- The intercalving period ranges **between 450-550 days**.

## Nagpuri (Or) Ellichpuri

- The breeding tract of this breed is Nagpur, Akola **and Amrawati districts of Maharashtra**.
- These are **black coloured animal with white patches** on face, legs and tail.
- This is also called as **Elitchpuri or Barari**.
- The horns are long, flat **and curved, bending backward** on each side of the back almost to shoulder (**sword shaped horns**).
- The **average milk yield is 700-1200 kg per lactation**.
- The age at **first calving is 45-50 months** with an **inter-calving period of 450-550 days**.



**Figure 7 – Nagpuri breed**

## Godavari

- Godavari is a result of **crossing of native buffaloes with Murrah bulls**. The home tract is Godavari and Krishna deltaic area
- The animals are of **medium stature with compact body**. The colour is predominantly black with a sparse coat of coarse brown hair.
- Godavari buffaloes are reputed for high fat with **daily average milk yield of 5-8 litres** and **lactation yield of 1200-1500 litres**.
- The animals breed regularly and have a **short calving interval** compared to Murrah.

## Toda

- Toda **breed of buffaloes** is named after an ancient tribe, **Toda of Nilgiris** of south India.



- Coat colour of the **calf is generally fawn at birth.**
- In adult the **predominate coat colours** are fawn and ash-grey.
- These buffaloes are quite distinct from other breeds and are indigenous to Nilgiri hills.
- The animals have **long body, deep and broad chest, and short and strong legs.**
- The head **is heavy with horns** set well apart, curving inward outward and forward.
- Thick **hair coat is found all over the body.** They are gregarious in nature.



**Figure 8 – Toda breed**

## Pandharpuri

- Native of **Kolhapur, Solapur districts** in south Maharashtra.
- **Body colour varies from light black to deep black.**
- It is **medium sized animal** having long narrow face, very prominent and straight nasal bone, comparatively narrow frontal bone and long compact body.
- Typical characteristic **of this breed is its horns which are very long**, curved backward, upward and usually twisted outwards. The horns are very long extending **beyond shoulder blade**, sometimes up to pin bones



**Figure 9 - Pandharpuri**

## Reproductive Details of Buffalo

- **Age at puberty** - 24-30 months
- **Age at first mating**-30-36 months
- **Optimum weight at first mating**-300-350 kg
- **Oestrus cycle length**-21 days
- **Duration of oestrus**-12 -18 hours
- **Time of ovulation**-12-16 hour after end of estrus
- **Optimum time of insemination**-Mid heat / towards the end of heat
- **Gestation period**-305-318 days
- **Dry period**-90 – 120

- **Calving to first heat**-40 days
- **Milk yield in litres**-1500-3000
- **Birth weight**-30-40 kg

# **Insemination** - insemination refers to the **process of introducing sperm cells into the female reproductive tract** to achieve pregnancy.

**There are two main types of insemination:**

**Natural insemination**: This is the **traditional method where a bull/bullock mounts a cow/buffalo** and deposits sperm cells during mating.

**Artificial insemination (AI)**: This is a **more controlled method where semen** (sperm cells and seminal fluid) **is collected from a bull, processed, and then deposited into the female reproductive tract** using specialized instruments. **AI is the most common method of insemination** in modern cattle breeding

## **COW BREEDS**

**In India two types of cattle breeds are domesticated exotic and indigenous breeds.**

**Exotic breeds** - Exotic cattle **breeds are breeds of cattle that are not native** to a particular region. They are typically imported from other **countries for the purpose of improving the local herd through crossbreeding.**

**Some of important characteristics**

- **High milk production**: Many exotic cattle breeds are **known for their exceptional milk yield**. This is a major reason why they are often crossbred with local breeds.
- **Large size**: Many exotic cattle breeds are larger than **their indigenous counterparts**.
- **Lower heat tolerance**: Because they are **not acclimated to hot climates**, some exotic cattle breeds may struggle in regions with high temperatures.
- **Lower disease resistance**: Similarly, **exotic cattle breeds may be more susceptible to diseases** that are not prevalent in their region of origin.

## Exotic Breeds of Cattle in India

| Breed                       | Origin                              | Characters                                                                                                                                                                                                                          |
|-----------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Jersey</b>               | Island of Jersey ( <b>England</b> ) | <ul style="list-style-type: none"> <li>➤ <b>Longest lactation</b> period among cattle (365 days)</li> <li>➤ <b>Highest milk fat</b> among cattle (5.5%),</li> <li>➤ <b>Smallest &amp; most hardy</b> exotic cattle breed</li> </ul> |
| <b>HF Holstein Friesian</b> | Netherland/Holland                  | <ul style="list-style-type: none"> <li>➤ <b>Highest milk</b> producer in world(6150 Liters/lactation)</li> <li>➤ <b>Lowest milk fat (3.5%)</b></li> </ul>                                                                           |
| <b>Ayrshire</b>             | Scotland                            | <ul style="list-style-type: none"> <li>➤ Most <b>beautiful</b> dairy breed</li> </ul>                                                                                                                                               |
| <b>Brown Swiss (BS)</b>     | Switzerland                         | <ul style="list-style-type: none"> <li>➤ Highest <b>milk lactose (5%)</b>,</li> <li>➤ The <b>maximum heat tolerance</b> exotic breed of cattle</li> </ul>                                                                           |



**Figure 10 - Jersey**



**Figure 11 - Holstein Friesian**





**Figure 12 - Ayrshire**



**Figure 13 - Brown Swiss**

## Indigenous Breed

Indigenous cattle breeds are **those that are native to a particular region**. They have **adapted to the local environment, climate, and feed resources** over generations. Here are some key characteristics of indigenous cattle breeds

### Key characteristics of indigenous cattle breeds

- **Well-adapted:** Indigenous cattle breeds are well-suited to the local climate, feed availability, and disease profile of their region. This makes them efficient and resilient animals.
- **Heat tolerance:** Because they have evolved in warm climates, indigenous cattle breeds are generally more heat tolerant than exotic breeds.
- **Disease resistance:** Indigenous cattle breeds have a natural resistance to diseases that are common in their region.
- **Lower milk yield:** While some indigenous breeds are good milk producers, they typically don't match the high milk volumes of exotic dairy breeds.
- **Multiple purposes:** Many indigenous breeds are dual-purpose breeds, meaning they are raised for both milk and meat production, or for draught work alongside milk or meat production.

## Indigenous Breed

**Indigenous Breeds are classified under three groups based on utility / purpose.**

- 1. Milch breed/milk breeds**
- 2. Dual purpose breeds**
- 3. Draught breeds**

## Classification of Indian cattle breeds

| Milch Breeds | Dual Purpose Breeds | Draught Breeds |
|--------------|---------------------|----------------|
| Gir          | Haryana             | Amritmahal     |
| Sahiwal      | Rathi               | Nagori         |
| Red Sindhi   | Kankrej             | Khillari       |
| Tharparkar   | Mewati              | Malvi          |
|              | <b>Dangi</b>        | Bachur         |
|              | Deoni               | Bargur         |
|              | Nimari              | Siri           |
|              | Ongole              | Ponwar         |
|              |                     | Hallikar       |
|              |                     | Kangayam       |

## **Milch Breeds / Milk Breeds:**

The cows of **these breeds are high milk yields and the male animals** are slow or poor work animals.

**The examples of Indian milch breeds are shahiwal, Red Sindhi, Gir and Deoni** the milk production of milk breeds is on the average more than 1600 kg per lactation.

## Red Sindhi

- Hailing from the Kohistan, Sindh **province in present Pakistan**, this breed is one of the most distinctive cattle breeds of india.
- Mainly available in **Punjab, Haryana, Karnataka, Tamil Nadu, Kerala and Orissa.**

- Under good management conditions **the Red Sindhi averages over 1700 kg of milk after suckling their calves but under optimum conditions there have been milk yields of over 3400 kg per lactation.**



**Figure 14 – Red sindhi**

## Sahiwal

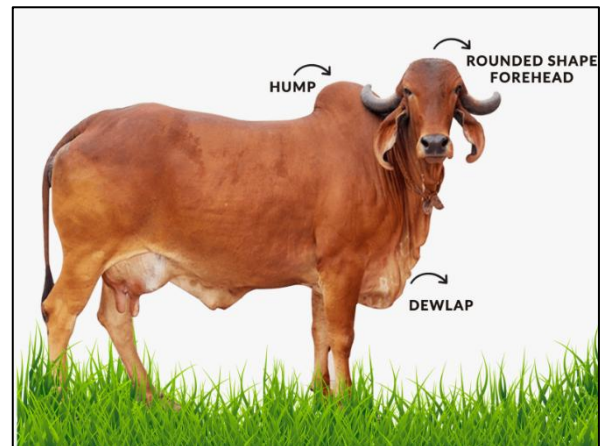
- Originally Belonging to the **Montgomery district of Present Pakistan**
- Mainly found in **Punjab, Haryana, U.P, Delhi, Bihar and M.P.**
- **Milk yield** – Under village condition :1350 kg
- **Milk yield** – Under commercial farms: 2100 kg
- Age at **first calving -32-36 months**
- **Calving interval** – 15 months



**Figure 15 – Sahiwal**

## Gir

- Mainly **found in Gir forest areas of South Kathiawar**
- **Gir Cows are good Milk – yielder**
- **Milk yield** – Under village **condition: 900 kg**
- **Milk yield** – Under commercial farms: **1600 kg**



**Figure 17 – Gir**

## Deoni

- Mainly **found in North western and western parts of A.P.**
- Cows are **good milk producers and bullocks** are good for work

## Dual Purpose Breeds:

The cows in these breeds are **average milk yielder and male animals are very useful for work.**

- Their milk production per **lactation is 500 kg to 150 kg.**
- **The example of this group is Ongole, Hariana, Kankrej, Tharparker, Krishna valley, Rath and Goal Mewathi.**

## Draught Breeds

The male animals are **good for work and Cows are poor milk yielder** are their milk yield as an **average is less than 500 kg per lactation.**

- They are **usually white in color.**
- A pair of bullocks **can haul 1000 kg.**
- **The example of this group Kangayam, Umblacherry, Amritmahal, Hallikar.**

## Milch And Draught Breeds

### **Haryana**

Mainly found in **Karnal, Hisar and Gurgaon district of Haryana, Delhi and Western M.P**

- **Milk yield** -1140 -4500 kgs
- **Bullocks are powerful for road transport and rapid ploughing**



**Figure 18 – Haryana**

### **Kankrej**



**Figure 19 - Kankrej**

- **Mainly found in Gujarat**
- **Milk yield** – Under village condition :1300 kg
- **Milk yield**– Under commercial farms: 3600 kg
- Age at **first calving -36 to 42 months**
- **Calving interval** – 15 to 16 months
- **Bullocks are fast, active and strong.** Good for plough and cart purpose

## Tharparkar

- **Mainly found in Jodhpur, Kutch and Jaisalmer**
- **Milk yield** – Under **village condition** :1660 kg
- **Milk yield** – Under **commercial farms**: 2500 kg



**Figure 20 - Tharparkar**

## Description Of Milch Breeds

| Breed              | Other names                              | Origin& Distribution                                 | Characters                                                                     |
|--------------------|------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Gir</b>         | Kathiawarhi, Surti                       | Gir forest (Guj.)                                    | <b>Longest lactation period among desi breeds (325 days)</b> , Best beef breed |
| <b>Sahiwal</b>     | Lola, Multani, Lambi-bar, Montgomery     | Montgomery(Pakistan), Punjab, Haryana, UP            | Best Indian milch breed, <b>Loose skin</b> (Lola)                              |
| <b>Red- Sindhi</b> | Red-Karachi                              | Karachi, Hyderabad (Sindh)                           |                                                                                |
| <b>Tharparkar</b>  | Malani, White Sindhi, Thari, Grey Sindhi | Tharparkar (Pakistan), Malani (Jaisalmer, Rajasthan) | <b>Highest disease resistant</b> , Colour change property                      |

## Description of dual- purpose breeds

| Breed               | Origin& Distribution                | Mixture                                | Characters                                                   |
|---------------------|-------------------------------------|----------------------------------------|--------------------------------------------------------------|
| <b>Hariana</b>      | Rohtak, Hisar<br>(Haryana)          |                                        | Best dual-purpose breed                                      |
| <b>Rathi</b>        | Alwar,<br>Bikaner (Rajasthan)       | Red Sindhi,<br>Sahiwal &<br>Tharparkar |                                                              |
| <b>Kankrej</b>      | Kutch ka Rann (Guj.)                |                                        | <b>Heaviest Indian Cattle breed Famous gait “Sawai Chal”</b> |
| <b>Mewati(Kosi)</b> | <b>Alwar, Bharatpur (Rajasthan)</b> | <b>Gir, Rathi&amp; Nagori</b>          |                                                              |
| <b>Dangi</b>        | Maharashtra, Gujarat                |                                        |                                                              |
| <b>Nimari</b>       | Narmada Valley (MP)                 | Gir, Khillari                          |                                                              |
| <b>Ongole</b>       | Ongole(AP)                          |                                        | <b>Cattle with pure white colour</b>                         |
| <b>Deoni</b>        | Hyderabad<br>(Telangana)            | Gir,Dangi & local animal               |                                                              |

## Description of draught- purpose breeds



| Breed                          | Origin                   | Characters                                  |
|--------------------------------|--------------------------|---------------------------------------------|
| <b>Amritmahal</b>              | Karnataka                | <b>Best draught- purpose breed of India</b> |
| <b>Nagori</b>                  | Rajasthan                | Best draught- purpose breed of Rajasthan    |
| <b>Khillari</b>                | Maharashtra              | Best draught- purpose breed of Maharashtra  |
| <b>Vechur</b>                  | <b>Kottayam (Kerala)</b> | <b>Smallest cattle breed in the World</b>   |
| <b>Bachaur</b>                 | Sitamari (Bihar)         |                                             |
| <b>Siri Sikkim, Darjeeling</b> | Sikkim, Darjeeling       |                                             |
| <b>Ponwar</b>                  | Ponwar Pilibhit(UP)      |                                             |
| <b>Kangayam</b>                | Tamilnadu                | Pride of Tamilnadu                          |
| <b>Bargur</b>                  | Coimbatore (Tamilnadu)   |                                             |
| <b>Hallikar</b>                | Karnataka                |                                             |
| <b>Malvi</b>                   | Malva (MP)               |                                             |

## Crossbreeds of cattle

### Cross breed –

**crossbreeding refers to the mating of animals from two or more distinct breeds** to produce offspring (calves) that inherit a mix of characteristics from both parents.

In crossbreeding generally **exotic breeds are mated with indigenous breeds** so that the offspring should have mix characteristics of both the parents.

| <u>Crossbred</u>    | <u>Cross</u>                |
|---------------------|-----------------------------|
| <b>Jersind</b>      | <b>Jersey × Red Sindhi</b>  |
| <b>Frieswal</b>     | HF×Sahiwal                  |
| <b>Jerthar</b>      | Jersey ×Tharparkar          |
| <b>Karan Swiss</b>  | Brown Swiss×Sahiwal         |
| <b>Karan Fries</b>  | HF × Tharparkar             |
| <b>Sunandini</b>    | Brown Swiss × non-descript  |
| <b>Jamaica Hope</b> | 80%Jersey, 15%Sahiwal, 5%HF |
| <b>Taylor</b>       | Short horn bull× Native cow |

**Taylor: first crossbred cattle developed in India**

## Important terms / Facts related to the Cattle

| <b>Particulars</b>                    | <b>Indigenous</b>                     | <b>Exotic / Crosses</b> |
|---------------------------------------|---------------------------------------|-------------------------|
| <b>Age at puberty</b>                 | 24 months                             | 12-15 months            |
| <b>Age at first mating</b>            | <b>30 months</b>                      | <b>18-20 months</b>     |
| <b>Optimum weight at first mating</b> | 250 kg                                | 180-275 kg              |
| <b>Oestrus cycle length</b>           | 17-24 days                            | 21±3                    |
| <b>Duration of oestrus</b>            | 12-18 hours                           | 12-18 hours             |
| <b>Time of ovulation</b>              | <b>12-16 hour after end of estrus</b> |                         |
| <b>Gestation period</b>               | <b>280-290 days</b>                   | <b>280-290 days</b>     |
| <b>Dry period</b>                     | 80 -90                                | 60 - 70                 |
| <b>Calving to first heat</b>          | <b>40 days</b>                        | <b>40 days</b>          |
| <b>Calving to first service</b>       | <b>60 days or less</b>                |                         |
| <b>Lactation length</b>               | 305                                   | 305                     |
| <b>Milk yield in litres</b>           | 1500-2000                             | 3500-5000               |

## Terminologies

**Puberty** - Puberty in **cattle is when a young animal reaches sexual maturity** and becomes capable of reproduction. This is a crucial stage in their development and affects breeding strategies in farms.

**Oestrus cycle** -The **estrous/ Oestrus cycle in cattle, also referred to as the heat cycle**, is the physiological process a female cow (heifer) goes through to prepare for pregnancy. It's a **recurring cycle that repeats itself approximately every 21 days (with a range of 18-24 days) and involves hormonal changes and physical signs.**

**Ovulation** - In cattle, ovulation is **a specific event within the estrous cycle** (also called the heat cycle) where a **mature egg is released from the cow's ovary**. It's a critical part of the reproductive process, as ovulation allows for fertilization by sperm to create an embryo.

**Gestation period** - The **gestation period in cattle, also known as pregnancy length**, varies depending on several factors but typically **falls within a range of 279 to 295 days.**

**Dry period** - The **dry period in cattle refers to the time between a cow's last milking during pregnancy and her subsequent calving**. It's a crucial stage for both the cow's health and milk production in the next lactation cycle.

**Calving** - Calving, also **referred to as parturition**, is the process **by which a cow gives birth to her calf**. It's a natural process, but close monitoring is crucial to ensure the health of both the mother and the newborn.

**First service** - In cattle, the **first service refers to the first time a female cow (heifer) is bred, either through natural mating with a bull or artificial insemination (AI)**. It's a crucial stage in a heifer's reproductive life and directly affects factors like calving age and overall herd productivity.

**Lactation length** - **Lactation length refers to the total period a cow produces milk after calving. It's a crucial factor in dairy farming**, influencing milk production efficiency and herd management practices.

**There are some exotic and Indian breeds of sheep's**

## **Crossbreds of sheep**

| Crossbred           | Cross                                        |
|---------------------|----------------------------------------------|
| Hissardale          | Bikaneri ewe (Magra) × Australian Merino ram |
| Avikalin            | Rambouillet ram×Malpura ewe                  |
| Avivastra           | Rambouillet ram×Chokla ewe                   |
| Indian Karakul      | Marwari,Malpura&Sonadi×Karakul               |
| Nilgiri Synthetic   | Nilgiri× Merino& Rambouillet                 |
| Patanwadi Synthetic | Patanwadi× Merino& Rambouillet               |



**Figure 21 - Hissardale**



**Figure 22 – Indian  
karakul**



**Figure 23 - Nilgiri**

## **Exotic sheep breeds**

- 1. Fine wool breeds:** -Merino, Rambouillet
- 2. Fur breeds:** -Karakul
- 3. Mutton breeds:** -Dorset, Suffolk, Southdown
- 4. Long coarse wool breeds:** -Lincoln, Leicester

## Description of exotic sheep breeds

| Breed              | Origin                                               | Characters                                                                        |
|--------------------|------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Merino</b>      | Spain                                                | <b>Best fine wool breed in the world, Ewes polled&amp; rams have spiral horns</b> |
| <b>Rambouillet</b> | Rambouillet (France)                                 | Developed from Spanish Merino                                                     |
| <b>Lincoln</b>     | England                                              | <b>Largest &amp; heaviest breed in the world</b>                                  |
| <b>Leicester</b>   | England                                              | Purest English breed                                                              |
| <b>Southdown</b>   | England                                              | Mousey grey face, Oldest English breed                                            |
| <b>Karakul</b>     | <b>Bukhara province of Uzbekistan (Central Asia)</b> | <b>Best Pelt (good quality fur) producing breed in the world</b>                  |
| <b>Suffolk</b>     | England                                              | Best mutton breed                                                                 |

## Description Of Indian Sheep Breeds

### 1. Chokla (Shekhawati)

**Origin** -Sikar, Churu & Jhunjhunu (Rajasthan)

#### Characters-

- Roman nose,
- Medium size compact body;
- Known as “**Indian Merino**”

**Figure 24 – Chokla**



### 2. Nali

**Origin-** Ganganagar, Churu & Jhunjhunu (Rajasthan)

**Characters** - Canary coloration of **wool (yellow wool)**

### 3. Marwari

**Origin** - Marwar reg. (Rajasthan)

**Characters-**

- **Highest disease resistant,**
- **Triple purpose sheep breed**

### 4. Pugal

**Origin**-Bikaner (Rajasthan)

**Imp characters** - yellow streak above eyebrow

### 5. Nellore

**Characters**

- **Tallest breed of sheep in India,**
- **Best mutton breed of India,**
- **Also known as “goat-like sheep” (No wool)**

### 6. Mandya (Bannur)

**Origin** - Karnataka

**Characters** - **Smallest sheep breed**

### 7. Garole

**Sundarban (WB)**

Most prolific **sheep breed of India**

### 8. Gaddi (nomadic tribe Gaddis rear this breed)

**Kullu & Kangra Valleys (HP)**

**Characters** - Very sturdy & very good climber

Polled ewes & horned rams,

Undercoat used for making “Kullu shawls” & blankets

### 9. Bhakarwal

**[www.crackgradeb.com](http://www.crackgradeb.com) | [info@crackgradeb.com](mailto:info@crackgradeb.com) | 9310558455**



**Figure 25 – Marwari**



**Figure 26 - Nellore**



- nomadic **tribe Bhakarwal** rear this breed
- **Lower hills of Himalaya**
- **Rams horned** but ewes hornless,
- **“Loei” is obtained from this breed**

## Classification Of Indian Sheep Breeds

| North temperate region breed | Northwestern arid & semi-arid region breed | Southern peninsular region breed | Eastern region Breed |
|------------------------------|--------------------------------------------|----------------------------------|----------------------|
| Gaddi                        | Chokla (Shekhawati)                        | Nellore                          | Chottanagpuri        |
| Bhakarwal                    | Nali                                       | Mandya                           | Garole               |
| Poonchi                      | Marwari                                    | Deccani                          | Ganjam               |
| Gurez                        | Jaisalmeri                                 | Hassan                           |                      |
| Rampur Bushair               | Pugal                                      | Bellary                          |                      |
| Karnah                       | Magra(Bikaneri)                            | Madras Red                       |                      |
| Changthangi                  | Malpura                                    |                                  |                      |

## Reproductive parameters of sheep

- **Breeding age-** 6-8 months
- **Comes to heat after lambing** – 21 days after
- **Length of pregnancy (gestation period)- 147 days** (ranges between 144 and 152 days)
- **Male female ratio** - 1:20 (1 male is sufficient for 20 females for breeding purpose)
- **Estrous period is repeated every 16-17 days** on average in ewes (range 14-19 days).
- **Estrous period is 19-21 days** in does (range 17-24 days).
- **The estrous period lasts for about 24-36 hours** in ewes and 34-38 hours in does.

| Sheep | Male adult -<br>Ram | Female adult -Ewe | Group of animals<br>-Flock | Act of mating<br>Tupping | Act of parturition<br>-Lambing | New born<br>- Lamb | Young male<br>Ram<br>lamb | Young female -<br>Gimmer<br>lamb |
|-------|---------------------|-------------------|----------------------------|--------------------------|--------------------------------|--------------------|---------------------------|----------------------------------|
|-------|---------------------|-------------------|----------------------------|--------------------------|--------------------------------|--------------------|---------------------------|----------------------------------|

## Goat Breeds

| Breed        | Origin & Distribution | Characters                                                                                                                      |
|--------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Saanen       | Switzerland           | Also known as “milk queen”                                                                                                      |
| Alpine       | Africa                |                                                                                                                                 |
| Angora       | Turkey                | “Mohair” (white soft silky hairs) producing breed of goat, Also known as “sheep like goat”                                      |
| Anglo-Nubian | England               | Roman nose, also known as “Jersey of goat” most outstanding dual-purpose goat breed widely used for the upgrading of desi stock |
| Toggenburg   | Switzerland           | Horn absent, maximum adaptability in exotic breeds                                                                              |
| Boer         | Southern Africa       | Meat type breed used for crossbreeding for meat                                                                                 |
| Orenberg     | Kazakhstan            |                                                                                                                                 |



**Figure 27 – Angora Goat**



**Figure 28 – Anglo – Nubian**

## Indian Goat Breeds

| North temperate region breed | Northwestern arid & semi-arid region breed | Southern peninsular region Breed | Eastern region Breed |
|------------------------------|--------------------------------------------|----------------------------------|----------------------|
| Chegu                        | <b>Jamunapari</b>                          | Malabari                         | <b>Black-Bengal</b>  |
| <b>Changthangi</b>           | Barbari                                    | Osmanabadi                       | Ganjam               |
| Gaddi                        | <b>Jhakrana</b>                            | <b>Sangamneri</b>                |                      |
|                              | Beetle                                     |                                  |                      |
|                              | Sirohi                                     |                                  |                      |
|                              | <b>Kutchi</b>                              |                                  |                      |
|                              | Marwari                                    |                                  |                      |

## Description of Indian goat breeds

### 1. Jamunapari –

- Etawah, **Agra (UP)**
- Biggest & most majestic breed of Indian goat,
- “**Roman nose**”(parrot mouth appearance)
- Most **beautiful Indian goat, Creamy white body colour** with shining hair,
- Thick growth of **hairs on buttocks known as “Feathers”**



**Figure 29 -  
Jamnapuri**

### 2. Barbari ( City breed)

- Etawah, Agra (UP), Bharatpur (Raj.)
- Origin: **Barbera city (East Africa) Dwarf breed highly** suitable for stall feeding & generally found in cities so **it is also known as “city breed” Indian goat with maximum milk fat (5%)**

### 3. Beetle

- **Punjab, Haryana**
- **Buck possesses marked beard but females** are beardless, roman nose

### 4. Jhakrana

- Alwar (Raj.)
- **Skin popular for tanning purpose**

## 5. Kutchi

- **Corkscrew** shaped horns

## 6. Black Bengal

- Nadia (WB)
- **Best chevon breed of India**, highly prolific goat breed of India

## 7. Chegu

- Lahul & Spiti valleys (HP)
- Pashmina producing breed

## 8. Changthangi (also known as Pashmina)

- Changthang (Ladakh)
- Pashmina producing breed, **Pashmina used for making high-quality Kashmiri “Rug or Shawl”**



**Figure 30- Changthani breed**

**Note: -**

**1. Chegu & Changthangi goat breeds produce Pashmina** while other breeds of North temperate region (Gaddi) produce medullated fibres.

**2. Pashmina:** fine costly hairs harvested from both Chegu & Changthangi (Goat breeds)

### Important terminologies

| <u>Animal</u> | <u>Adult male</u> | <u>Adult female</u> | <u>Group of animals</u> | <u>Act of mating</u> | <u>Act of parturition</u> | <u>New born</u> | <u>Young male</u> | <u>Young female</u> |
|---------------|-------------------|---------------------|-------------------------|----------------------|---------------------------|-----------------|-------------------|---------------------|
| <b>Goat</b>   | Buck              | Doe/nanny           | Trip or band            | Serving              | Kidding                   | Kid             | Buckling          | Gloating            |

### Reproductive Details of Goats

- **Doe may be mated when 10 to 15 months old** so that they kid at the age of 15 to 20 months.
- But as a rule, a goat should not be mated **until it is one year old. It is better to breed the female once a year**
- The **average gestation period is 151 ± 3 days**.
- Some goats can be made to **kid twice in 18 months**.

- The goats reach their maximum **efficiency at the age of five to seven years**.
- In exceptional cases they **continue to be serviceable even up to 12 years** and in rare **cases up to 14 years**.
- A **well-maintained doe may continue to be milked until a month before she is expected to kid again**.
- The condition of the doe during gestation will have a very great influence on the quality of kids at birth.
- A **doe in good condition will produce strong lively kids**, whereas a doe in poor condition may produce ungainly kids, weak in constitution.
- **Does must be fed well**, allowed liberal exercise and protected from rain and cold.

## Pig Breeds

### Indian Breeds of Pig

| <u>Breed</u>     | <u>Home tract</u>  |
|------------------|--------------------|
| <b>Ghoongroo</b> | <b>West Bengal</b> |
| Niang Megha      | Meghalaya          |
| Agonda goan      | Goa                |
| <b>Tenyi Vo</b>  | <b>Nagaland</b>    |
| Nicobari         | Andaman&Nicobar    |
| Doom             | Assam              |

### Exotic breeds of pig

1. **England breeds** -LWY, MWY, Berkshire, Tamworth
2. **Denmark breeds**: -Landrace
3. **U.S.A. breeds**: - Duroc, Chester-White, Poland-China

### Description Of Exotic Breeds of Pig

| Breed                        | Home tract | Characters                                                                                           |
|------------------------------|------------|------------------------------------------------------------------------------------------------------|
| Large White Yorkshire (LWY)  | England    | Heaviest breed of pig                                                                                |
| Middle White Yorkshire (MWY) | England    | Excellent pork breed                                                                                 |
| Berkshire                    | England    | Black colour with 6 white marks on body                                                              |
| Tamworth                     | England    | Golden red colour body, Large, long & tall breed known for lean meat                                 |
| Landrace                     | Denmark    | Highest quality bacon in the world White colour breed with some black skin spots known as "Freckles" |

## Important terminologies

| <u>Animal</u> | <u>Adult male</u> | <u>Adult female</u> | <u>Group of animals</u> | <u>Act of mating</u> | <u>Act of parturition</u> | <u>New born</u> | <u>Young male</u> | <u>Young female</u> |
|---------------|-------------------|---------------------|-------------------------|----------------------|---------------------------|-----------------|-------------------|---------------------|
| Pig           | Boar              | Sow                 | Drove                   | Coupling             | Farrowing                 | Piglet          | Boarling          | Gilt                |

## Reproductive Details of Pig / Pig Breeding

- Pigs are **highly prolific in nature and two farrowing in a year** should be planned by adopting optimal management conditions
- A sow can **be breed as early as 8-9 months of age** and **can farrow twice in a year**. They **produce 6-12 piglets in each farrowing**.
- **Male piglets not selected for breeding should be castrated preferably at the age of 3-4 weeks** which will prevent the boar odour in the cooked meat thus it enables production of quality meat.
- **For every 10 sows one boar** must be maintained for maximum fertility.
- **Heat period of the sow is once in about 21 days**
- Breed the animals when it is in **peak heat period (i.e. 12 to 24 hours of heat)**.



- **Gestation period of pig is three months**, three weeks and three days. **That is 115 days, with a range of 111 to 120 days**, from the first day of mating.
- **Farrowing-** Farrowing refers to the **act of a sow, a female pig, giving birth (Parturition) to a litter of piglets**. It can also refer to the **litter itself**. **Only two farrowing's should be performed in a year.**
- **Cull the old animals after 10-12 farrowing's.**
- **Weaning period** (separation of the piglet from the mother)– **after 2 months**
- **Uncastrated males and females should not be housed together beyond the age of four months.**

### Some important information about piggery

- The pig has **got highest feed conversion efficiency** i.e. they produce more live weight gain from a given weight of feed than any other class of meat producing animals except broilers.
- The **pig can utilise wide variety of feed stuffs viz. grains, forages, damaged feeds and garbage and convert them into valuable nutritious meat**. However, feeding of damaged grains, garbage and other unbalanced rations may result in lower feed efficiency.
- Pig farming **requires small investment** on buildings and equipment's
- **Pigs are known for their meat yield, which in terms of dressing percentage ranges from 65 - 80 in comparison** to other livestock species whose dressing yields may not exceed 65%.
- **Pigs manure is widely used as fertilizer for agriculture farms and fish ponds.**
- **Pigs store fat rapidly for** which there is an increasing demand from **poultry feed, soap, paints and other chemical industries.**
- Pig farming provides quick returns since the marketable weight of fatteners can be **achieved with in a period of 6-8 months.**
- **There is good demand from domestic as well as export market** for pig products such as pork, bacon, ham, sausages, lard etc

## Breeds Of Horses & Ponies

### Indian horse breeds

| Breed      | Home tract         | Characters                                                                           |
|------------|--------------------|--------------------------------------------------------------------------------------|
| Kathiawari | Kathiawar (Guj.)   | Best breed of horse                                                                  |
| Manipuri   | Manipur            | Manipuri horses are known for their intelligence so used for polo, military & racing |
| Marwari    | Marwar (Raj.)      | These horse breeds are named after the place they are found                          |
| Spiti      | Himachal Pradesh   |                                                                                      |
| Zanskari   | Leh & Ladakh (J&K) |                                                                                      |
| Bhutia     | Sikkim, Darjeeling |                                                                                      |

## Camel Breeds

**Breeds of Rajasthan:** - Bikaneri, Jaisalmeri, Mewari, Mewati, Marwari, Jalori

**Breeds of Gujarat:** - Kutchi, Kharai

**Breeds of MP:** - Malvi

**Kharai**- Also known as “**swimming camel**” It can **swim up to 3 Km.** in to the sea in search of mangrove (primary food) Tolerant to high saline water

**Note:** -

- **Indian camels are mainly two types**- Plain camels & Hill camels
- **Plain camels include**- Riverine camels & Desert camels
- **Hill camels & Riverine camels are Baggage type** whereas Desert camels are Riding type
- **Desert camel breeds are Bikaneri & Jaisalmeri**

## Housing system of animals

Animal housing systems are designed to **provide animals with a safe, comfortable, and healthy environment**. The specific type of housing system used will vary depending on the species of animal, the climate, and the production goals.

**There are two main types of housing systems for animals: confinement housing and loose housing.**

- 1. Conventional dairy barn/ Stanchion dairy barn** -is where **animals are kept in stalls or cages**. This type of housing is often used for intensive production systems, such as factory farms, dairy farms etc.
- 2. Loose housing is where animals are allowed to move around freely** within a building or enclosure. This type of housing **is more welfare-friendly** than confinement housing and is becoming increasingly popular.

### The Requirement of Loose Housing System

| Types of animals | Floor space per animal for Covered area (Sq. feet) | Floor space per animal for Open area (Sq. feet) |
|------------------|----------------------------------------------------|-------------------------------------------------|
| Young stock      | 15-20                                              | 50-60                                           |
| Cows             | 20-30                                              | 80-100                                          |
| Buffaloes        | 25-35                                              | 80-100                                          |
| Pregnant cows    | 100-120                                            | 180-200                                         |
| Bulls            | 120-140                                            | 200-250                                         |

### Requirements of loose housing system as per BIS

| Types of animals | Floor space per animal for Covered area (m <sup>2</sup> ) | Floor space per animal for Open area (m <sup>2</sup> ) |
|------------------|-----------------------------------------------------------|--------------------------------------------------------|
| Young calves     | 1.0                                                       | 2.0                                                    |
| Cows             | 3.5                                                       | 7.0                                                    |
| Bullocks         | 3.5                                                       | 7.0                                                    |
| Buffaloes        | 4.0                                                       | 8.0                                                    |
| Bulls            | 12                                                        | 120                                                    |

## Conventional Dairy Barn

- Cow-shed
- Calving box (100-150 Sq.ft)
- Isolation box (150 Sq.ft)
- Young stock shed
- Bull shed (150 Sq.ft)

## Housing Requirement for Pigs

**Housing of boars (male pig)** - Boar pen **should have covered area of 6-7 m<sup>2</sup> and open area of 8-12 m<sup>2</sup> for exercise.** The walls should have a minimum height of 1.5 m.

**Housing of Sow (female pig)** - Open yard type with partial roofing as in the case of boar may be provided. **A total of 10-15 females can be grouped in a pen. An area of 2 m<sup>2</sup> per animal may be provided.**

**Housing of Farrowing sows** - Farrowing sows may be housed individually in a farrowing pen of 2.5 x 4.0 = 10.0 m<sup>2</sup> having guard rails, creep area, feed and water troughs.

**Housing of growing and finishing pigs**- A covered concrete yard for feeding and resting having feed and **water trough arranged in the front side and an open yard** in the rear will suffice for fatteners. **The total space requirement may be 2 m<sup>2</sup> per grower/fattener pig.**

## Housing Requirement for Sheep and Goat

### Floor space requirement per animal (sheep/ goat) (BIS standard)

| Types of animals         | Minimum floor space per animal (m <sup>2</sup> ) |
|--------------------------|--------------------------------------------------|
| Ram or buck in groups    | 1.8                                              |
| Ram or buck - individual | 3.2                                              |
| Lambs or kids - in group | 0.4                                              |
| Weaner in groups         | 0.8                                              |
| yearling or goatlings    | <b>0.9</b>                                       |
| Ewe or doe in groups     | 1.0                                              |
| Ewe with lamb            | 1.5                                              |

## Average life span & age at puberty

| Animal       | Average life span    | Onset of Puberty            |
|--------------|----------------------|-----------------------------|
| Canine       | 10 years             | 7-10 months                 |
| Swine        | 6-10 years (8 years) | 6-7 months                  |
| Sheep        | <b>12 years</b>      | <b>7-10 months</b>          |
| Goat         | 15 years             | 7-10 months                 |
| Horse        | <b>20 years</b>      | <b>18 months (1.5 year)</b> |
| Dairy cattle | 20 years             | 12 months                   |
| Zebu cattle  | <b>20 years</b>      | <b>18-24 months</b>         |
| Buffalo      | 20-25 years          | 18-24 months                |

## Age of castration & weaning

**Castration** -Castration refers to **the removal or inactivation of an animal's reproductive** organs, most commonly the testes in males and ovaries in female.

## Reasons for Castrating Animals

**Pet population control:** Castration helps to reduce pet homelessness and overpopulation.

**Behaviour modification:** Castration can improve undesirable behaviours in male animals, such as aggression, mounting, and marking territory with urine.

**Health benefits:** As mentioned earlier, castration can offer some health benefits for certain animals.

## Methods Of Castration

**A. Open method/surgical method-** Removal of **both testicles** through incision

**B. Closed method-**

**1. Burdizzo castrator:**

- It **separately crushes** each spermatic cord.
- It is a **bloodless castration**.
- It is the best **method of castration**.

**2. By Rubber ring method**

## Special points

- **Castration in female is known as spaying or ovariectomy** (removal of both ovaries & uterus)
- Castrated female is known as *Spayed*
- Castrated fowl is **known as Capon & process is caponisation**
- **Castrated Tom (male cat)** is known as Neuter & process is neutering

**Weaning:** the **process of separating the baby from its mother** at a very early age, up to this age whole milk feeding is desired.

**Note:** Early weaning is difficult in buffaloes due to high motherly instinct.

## Weaning is Important because-

**Nutritional Needs:** As young animals grow, their nutritional needs change. **Milk can no longer provide all the nutrients they require for proper development.** Solid food offers a broader range of essential nutrients.

**Digestive Development:** The digestive system of a young animal matures as they start consuming solid food. Weaning helps stimulate this development and allows the digestive system to function efficiently with a new food source.

**Behavioural Independence:** **Weaning encourages young animals** to become more independent from their mothers. They **learn to forage for food on their own**, which is crucial for survival in the wild.

| Animal | Age of castration | Age of weaning       |
|--------|-------------------|----------------------|
| Calf   | 8-10 weeks        | 90 days (8-12 weeks) |
| Lamb   | 2 weeks           | 14 weeks             |
| Kid    | 2-4 weeks         | 16 weeks             |
| Piglet | 1 week            | 8 weeks              |

**Dehorning/Disbudding:** - the process of **removing horns/horn buds of an animal** after birth.



**Best age of dehorning in calf: before 10th day old (up to this time horn button does not attach to the skull)**

**Methods of dehorning: -**

- 1. Chemical method** - KOH (caustic potash) is used
- 2. Electrical method** - Electrical dehorner (10000 F/10 seconds)
- 3. Mechanical**-by clipper saws or by rubber bands

**Dairy Science**

**Standards for SNF according to Prevention of Food Adulteration Act (PFA-1976)**

| <b>Solid Not Fat (SNF)</b> | <b>Animals milk</b>                                           |
|----------------------------|---------------------------------------------------------------|
| <b>9.0%</b>                | <b>Buffalo milk, Sheep milk, Goat milk, Double toned milk</b> |
| <b>8.7 %</b>               | Skim milk                                                     |
| <b>8.5 %</b>               | Cow milk, Toned milk, Standard milk, Recombined milk          |

**SNF (Solid not fat)** -SNF refers to the portion of milk that contains all the components except water and fat. This includes protein, lactose (milk sugar), and minerals like calcium and phosphorus. A higher SNF content in milk generally indicates a higher nutritional value.

**Standards for milk fat according to PFA-1976**

| <b>Milk fat</b> | <b>Animals milk</b>                                |
|-----------------|----------------------------------------------------|
| <b>0.5 %</b>    | Skim milk                                          |
| <b>1.5 %</b>    | <b>Double toned milk</b>                           |
| <b>3.0 %</b>    | Toned milk, Sheep milk, Goat milk, Recombined milk |
| <b>3.5 %</b>    | <b>3.5 % Cow milk</b>                              |
| <b>4.5 %</b>    | Standard milk                                      |
| <b>5.0 %</b>    | Buffalo milk                                       |

## Chemical Composition of Milk Of Different Animals

| Animal  | Water | Fat | Protein |
|---------|-------|-----|---------|
| Cow     | 86.6  | 4.6 | 3.4     |
| Buffalo | 84.2  | 6.6 | 3.9     |
| Sheep   | 79.4  | 8.6 | 6.7     |
| Goat    | 86.5  | 4.5 | 3.5     |
| Camel   | 86.5  | 3.1 | 4.0     |

## Methods Of Pasteurization (For the Preservation Of Milk)

**Pasteurization** - Pasteurization of milk is the process of applying heat to kill harmful **bacteria** that can cause foodborne illnesses. It's named after Louis Pasteur, the scientist who discovered this method in the 1860s.

Milk pasteurization **is a controlled heating treatment that destroys pathogens** (disease-causing microorganisms) in milk by disrupting their cell walls and proteins. This significantly **reduces the risk of illnesses like typhoid fever, tuberculosis, and E. coli infections**.

| Methods                                                                | Temperature-time protocol              |
|------------------------------------------------------------------------|----------------------------------------|
| Low-Temperature-Long Time (LTLT) method /Batch/Holding pasteurization  | 63°C / 30 minutes                      |
| High-Temperature-Short-Time (HTST) pasteurization/Flash pasteurization | 72°C / 15 seconds                      |
| Stassanization                                                         | 74°C / 7 seconds                       |
| Ultra-high temperature (UHT) pasteurization                            | 130-135°C / fraction of second         |
| Ultra-pasteurization (Uperization)                                     | 150°C / fraction of second             |
| Ultra-pasteurization (Uperization)                                     | 150°C / fraction of second             |
| Vacuum pasteurization (Vacreation)                                     | By Vacreator instrument (direct steam) |

## Milking Methods

**A. Hand Milking**- most common practice **among individual farmers** it involves fisting, knuckling and stripping

- **Full hand method/fisting**- best method of milking dairy animals
- **Knuckling**- Wrong method of milking (injury may occur in teat)
- **Stripping**- It is used in following conditions:
  - (a) To **remove left milk** after machine milking
  - (b) When teats are too small
  - (c) Towards completion of milking

**B. Machine Milking**- practiced in organized government & military dairy farm

**It performs two basic functions: -**

- (a) It opens **the streak canal with partial vacuum**, allowing the continuous flow of milk
- (b) It **massages the teats that prevent congestion** of blood in the teat

## Colostrum/1st milk/Beestings: -

**Colostrum milk** - Colostrum, **also sometimes called beestings in** the case of cows and other mammals, is the **first milk produced by the mammary glands** following childbirth. **It's a thick, sticky, and yellowish liquid packed with nutrients specifically designed to give newborns the best possible start in life.**

- Colostrum is fed to **newborn calf @ 10 % of its body wt. (or) 2-2.5 Liters per day** up to 3-4 days.
- Milk lost its **colostral property within 3-4 days** after parturition.

## Meat Science

### The composition of animal meat

**72% water & remaining 28% dry matter (DM)** (DM consists of 21% Protein, 6% Fat & 1% Ash)

- **Bright red color of meat is due to oxymyoglobin**

- **Brown color of meat** is due to metmyoglobin
- **Pink color of cured meat** is due to nitro-haemochromogen
- **Variety meat/organ meat:** -organs such as tongue, brain, thymus, heart, liver & kidney are known as variety meat
- **Pork contains 5-10 times more vita. B1** (thiamine) than beef & mutton.
- **Folic acid & vita. B12 is higher in beef & mutton**
- **Bacon:** meat from the sides and the back of a pig
- **Ham: meat of the thigh or buttocks** of a pig
- **Prime pork:** carcass of castrated male & virgin female of a pig
- **Poultry meat is known as white meat** while the meat of other animals is known as red meat
- **Vita.B12:** Vitamin found in highest amount in animal meat.
- **Niacin:** Vitamin found in highest amount in poultry meat.
- **Vitamin A is present in beef and mutton but absent in buffalo meat, chevon or pork**

| Animal  | Meat               | Color of meat     |
|---------|--------------------|-------------------|
| Goat    | Chevon             | Light red         |
| Sheep   | Mutton             | Light to dark red |
| Cattle  | Beef               | Cherry red        |
| Buffalo | Cara-beef/ Buffen  | Cherry red        |
| Pig     | Pork               | Greyish pink      |
| Poultry | White meat/Chicken | Grey-white        |
| Calf    | Veal               | Brownish pink     |

## Animal Reproduction

### Estrous cycle & estrus phase (heat period) of various animals

**Estrous Cycle** - The estrous cycle in animals is a **series of recurring physiological changes in the female reproductive system** that prepares them for mating and pregnancy.

**Estrus:** This is the "**heat**" period when the female is receptive to mating. High estrogen levels **trigger behavioural changes to attract mates** (increased vocalizations, mounting behaviour). Ovulation (egg release) typically occurs during this stage.

| Animal           | Length of estrous cycle | Length of estrus phase (heat period) |
|------------------|-------------------------|--------------------------------------|
| <b>Cattle</b>    | <b>21 days</b>          | 12-24 hours                          |
| <b>Mare</b>      | 21 days                 | 6 days                               |
| <b>Sow</b>       | 21 days                 | 2-2.5 days                           |
| <b>She-camel</b> | 23 days                 | 3-5 days                             |
| <b>Ewe</b>       | <b>17 days</b>          | 36 hours                             |
| <b>Doe</b>       | 19 days                 | 48 hours                             |

**Ewe** -Ewe refers to a mature female sheep. Sheep are domesticated, ruminant mammals valued for their wool and meat.

**Mare** - A mare is an adult female horse.

**Doe** - A doe is the adult female of various mammals, most commonly referring to deer, rabbits, and kangaroos.

### The Gestation Period Of Common Domestic Animals

**Gestation period** -The gestation period of an animal refers to the **length of time a female carries fertilized eggs within her body until birth**. It's essentially the pregnancy duration for animals.

| Animal         | Gestation period                       |
|----------------|----------------------------------------|
| Cattle         | 283 days (9 month+ 9days)              |
| <b>Buffalo</b> | <b>310 days (10months+10days)</b>      |
| <b>Sheep</b>   | <b>145 days (5 months-5days)</b>       |
| Goat           | 155 days (5months+5days)               |
| <b>Sow</b>     | <b>114 days (3months+3weeks+3days)</b> |
| Mare           | 341 days (11months+11days)             |
| She-camel      | 390 days (1year+1month+1day)           |

## Animal Nutrition

**Cattle nutrition is just as important as nutrition for any other animal.** Like us, they need a balanced diet to **function properly, maintain good health, and be productive.**

**The specific nutrients and amounts required will vary depending on the cattle's:**

- **Age:** Calves, **growing heifers, adult cows, and bulls all have** different nutritional needs.
- **Physiological state:** **Pregnant and lactating cows** will need more nutrients than those simply maintaining their body weight.
- **Activity level:** Cattle that **are working animals or grazing on poor-quality** pastures will have higher energy requirements.
- **Breed:** Different cattle breeds may have slightly different nutritional needs.

| Item             | Hay                                    | Silage                                                                             |
|------------------|----------------------------------------|------------------------------------------------------------------------------------|
| Define           | Sun dried forage                       | Method of preserving green fodder for a long period of time <b>by fermentation</b> |
| Moisture content | 12-14%                                 | 60-70%                                                                             |
| Best crop        | Oat (Avena sativa)                     | Maize (Zea mays)                                                                   |
| Preferred plant  | Leafy Plant with thin stem             | Plant with thick stem                                                              |
| Harvesting time  | Flowering stages<br>(1/3 to ½ blossom) | Between flowering and milk stage                                                   |

### Hay

- Best hay is **prepared with Oat (Avena sativa)**
- Best legume hay is **prepared with Lucerne (alfa-alfa)**

### Advantages of haymaking:

- Sorghum (Jowar) **lost its cyanide toxicity during drying due to denaturation of** enzymes.



- **Vitamin D content is very high** in hay due to sun drying.

### Disadvantages of haymaking:

- **Vitamin A content of forage is lost** up to 80% due to sun drying.
- If **hay made with high moisture legume**, it results in fermentation & heat production leads to the formation of Brown hay or mow-burnt hay.
- **Nutritive value of hay is less** than that of silage.

### Silage

**Colour of good silage:** Yellowish or Brownish-green

**Haylage:** When grass & legume which are meant for hay are ensiled

- **Moisture** content of haylage is **45-55%**
- Having characteristics of both hay & silage

**Wastelage:** Anaerobically fermented animal waste (poultry dropping, poultry litter, swine excreta, and bovine dung) with the help of lactic acid producing bacteria

### Special points about silage making: -

- **Crops rich in soluble sugar such as maize, sorghum, bajra** are most suitable for ensiling.
- **Crops with less soluble carbohydrate** can also be used for ensilage by the addition of 3-3.5% molasses.
- **Mixture of grass/cereal fodder & legume used for ensilage should be in the ratio of 3:1**
- Urea @1% & salt @0.5% are added to cereals & grasses in order to improve N<sub>2</sub> content & palatability
- **Pit silos are more common in India.**
- **Flieg-index is commonly** used method for evaluation of silage quality.

## Difference between Roughage & Concentrate

| Items                                   | Roughage                                                                                                                   | Concentrate                                                                                                                                                                                                         |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Crude Fibre (CF)</b>                 | CF > 18%                                                                                                                   | CF < 18%                                                                                                                                                                                                            |
| <b>Total digestible nutrients (TDN)</b> | TDN < 60%                                                                                                                  | TDN > 60%                                                                                                                                                                                                           |
| <b>Energy content</b>                   | Low                                                                                                                        | High                                                                                                                                                                                                                |
| <b>Digestibility</b>                    | Low                                                                                                                        | High                                                                                                                                                                                                                |
| <b>Density of feed</b>                  | Low                                                                                                                        | High                                                                                                                                                                                                                |
| <b>Main function</b>                    | Belly-filling                                                                                                              | To provide energy                                                                                                                                                                                                   |
| <b>Examples</b>                         | a) <b>Dry roughage:</b> - Straw, stover, hay<br>b) <b>Succulent roughage:</b> - Silage, tree leaves, pasture, green fodder | A) <b>Energy-rich concentrate:</b> - Grains, seeds, brans, molasses<br>b) <b>Protein rich concentrate:</b> - Animal origin (fish meal, meat meal, feather meal, blood meal)<br>Plant origin (oil seed cakes, yeast) |

## Feeding Requirements

### (A) Dry matter requirement (per animal per day): -

- **Zebu cattle:** 2- 2.5 Kg/100 Kg body weight
- **Buffalo and crossbred cow:** 2.5 - 3Kg/100 Kg body weight
- **Sheep:** 2.5 - 3 Kg per animal per day
- **Lactating sheep:** 4% of body weight
- **Meat goat:** 3-4% of their live weight
- **Dairy goat:** 5-7% of their live weight
- **Camel:** 2% of body weight
- **Working & lactating camel:** 3% of body weight

### (B) Production ration: -

- **For Zebu Cattle:** 1Kg concentrate/2.5 Kg milk
- **For Buffalo & Cross-bred cow:** 1Kg concentrate/ 2 Kg milk
- **Gestation ration:** during last trimester of pregnancy
- **For Zebu Cattle:** 1.25 Kg concentrate
- **For Buffalo & Cross-bred cow:** 1.75 Kg concentrate

**Note: - Salt and mineral mixture should be given @ 1% of the concentrate mixture**

### Water requirement of domestic animals

**Water requirement is directly proportional to dry matter intake**

| Animal              | Drinking water requirement per animal per day |
|---------------------|-----------------------------------------------|
| Sheep & Goat        | 18 litres                                     |
| Poultry (100 birds) | 25 litres (250 ml per bird)                   |
| Pig                 | 22-25 litres                                  |
| Horse               | 36 litres                                     |
| Dairy Cattle        | 30-35 litres                                  |
| Buffalo             | 30-35 litres                                  |